

Exam Date & Time: 20-Nov-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VII Semester - End Semester Examination - Nov / Dec -2024

Machine Learning [IT-4032]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Compare and contrast supervised and unsupervised learning. Provide examples of real-world tasks that fall under each category and discuss how machine learning techniques are applied in these scenarios. (5)
 - A)
 - B) Evaluate the ethical implications of using ensemble learning methods such as bagging and boosting in decision-making systems. Discuss potential issues related to fairness, transparency, and accountability, and propose strategies for addressing these concerns. (5)
- 2) Compare the robustness of K-Means clustering, the EM algorithm, and Hierarchical clustering in handling overlapping clusters and noisy data. Which method is most resilient in scenarios with high variance in cluster density, and why? Provide examples of application scenarios where each method might excel or struggle. (5)
 - A)
 - B) Make use of strategies for tuning kernel parameters, such as the degree of polynomial kernels or the bandwidth of Gaussian RBF kernels, to optimize SVM model performance? How do these parameter choices impact model robustness and generalization across different datasets. (5)
- 3) Evaluate the main steps of the K-means clustering algorithm, examining processes such as initialization, cluster assignment, and centroid updating. (5)
 - A)
 - B) Compare parametric and non-parametric models in terms of their assumptions, flexibility, and scalability in classification and regression tasks. Provide examples where a parametric model like logistic regression might outperform a non-parametric model, and vice versa. Discuss the trade-offs involved in terms of model interpretability and generalization. (5)
- 4) The design of the fitness function in a genetic algorithm significantly impacts the algorithm's performance. Discuss how a poorly designed fitness function can lead to suboptimal solutions or convergence to local minima. How can knowledge from neural network loss functions be applied to improve fitness function design in genetic algorithms (5)
 - A)
 - B) Discuss the role of different distance metrics (e.g., Euclidean, Manhattan, Minkowski) in the K-NN algorithm. How does the choice of distance metric affect K-NN's (5)

performance, particularly in high-dimensional spaces. Provide examples of scenarios where each metric might be more suitable and explain how feature scaling impacts these choices

- 5) Evaluate the impact of data imbalance on the performance of classification algorithms such as logistic regression and support vector machines. Discuss techniques for handling imbalanced datasets and their implications for model fairness and interpretability. (5)

A)

B)

Consider the tuple $X = (\text{Weather} = \text{Rainy}, \text{Temperature} = \text{Mild}, \text{Humidity} = \text{High}, \text{Windy} = \text{True})$. Calculate the conditional probabilities required for predicting the class label 'PlayTennis' for both 'Yes' and 'No'.

Weather	Temperature	Humidity	Windy	PlayTennis
Sunny	Hot	High	False	No
Sunny	Hot	High	True	No
Overcast	Hot	High	False	Yes
Rainy	Mild	High	False	Yes
Rainy	Cool	Normal	False	Yes
Rainy	Cool	Normal	True	No
Overcast	Cool	Normal	True	Yes
Sunny	Mild	High	False	No
Sunny	Cool	Normal	False	Yes
Rainy	Mild	Normal	False	Yes

(5)

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